

# Reference Sheet for a Thesis with L<sup>A</sup>T<sub>E</sub>X2e and KOMA-Script

- All examples were tested with pdflatex.
- The package mentioned in the headings has to be included (see B.2).
- Compile three times after last change (esp. docs with references).

## A. L<sup>A</sup>T<sub>E</sub>X Basics

### A.1. Units

- Available units for length and dimensions:  
bp point (typographic) mm millimeter in inch em width of M  
px pixel (1/72in) cm centimeter pc pica ex height of x
- Document dependent units z\textwidth, z\linewidth, z\columnwidth, z\textheight with z a percentage value, e.g. 0.55\textwidth means 55% of the actual width of the text.
- \baselineskip minimum vertical space between the bottom of two successive lines in a paragraph.
- Amounts like \smallskipamount, \medskipamount, \bigskipamount.

### A.2. Reserved Characters (see also E.2, cf. H)

|     |                                                                                  |                |
|-----|----------------------------------------------------------------------------------|----------------|
| \   | introduces a command                                                             | \textbackslash |
| { } | embraces arguments, creates logical parts                                        | \{ \}          |
| [ ] | embraces <i>optional</i> arguments                                               | [ ]            |
| %   | comments: code after % will be ignored.                                          | \%             |
| &   | separates columns in tabular-like environments                                   | \&             |
| #   | parameter for own command declarations                                           | \#             |
| \$  | text style math mode (abbr. for \(...\))                                         | \\$            |
| ^   | index/exponent only valid in math mode, e.g. a <sub>1</sub> <sup>2</sup> see E.2 |                |

## B. Preamble (before \begin{document})

### B.1. Documentclass (necessary)

Use: \documentclass[opt,opt,...]{class}  
Recommended classes: scrartcl, scrreprt, scrbook, scrllttr2  
Non-KOMA-Script classes: beamer, koma-moderncvclassic

| Common options with default        | Values available (subtotal)          |
|------------------------------------|--------------------------------------|
| fontsize=11pt                      | 10pt   12pt (e.g. 12.5pt also valid) |
| paper=a4, paper=portrait           | a3   a5   b4   letter, landscape     |
| parskip=no                         | half   full                          |
| headings=big                       | small   normal                       |
| chapterprefix=false                | true                                 |
| open=right (scrbook)               | any (scrartcl, scrreprt)   left      |
| captions=oneline                   | nooneline                            |
| captions=tablebelow,figurebelow    | tableabove, figureabove              |
| toc=notlistof                      | listof   listofnumbered              |
| bibliography=totoc   totocnumbered | nottotoc                             |
| twoside=true (scrbook)             | false (scrartcl, scrreprt)           |
| twocolumn=false                    | true                                 |
| draft=false                        | true (show overfull boxes)           |

→ Options of document class are passed to every loaded package.  
→ Set or change options later in file, e.g. \KOMAoptions[twoside=true]

### B.2. Loading Packages

\usepackage[options]{package}  
\PassOptionsToPackage[options]{package}

### B.3. Encoding Settings

```
\usepackage[utf8]{inputenc} % most IDEs use UTF8
\usepackage[T1]{fontenc} % most fonts needs T1
```

### B.4. Language Settings with babel

Load: \usepackage[ngerman, main=english]{babel}  
Use: \selectlanguage{language} \foreignlanguage{language}{text}

```
\documentclass[italian]{scrbook} % global option
\usepackage[british,main=italian]{babel} % package option
\usepackage{csquotes} % package csquotes knows italian
```

## C. Layout

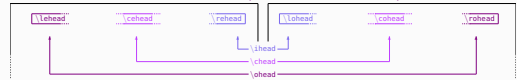
### C.1. Changing Page Layout with geometry

- Let KOMA-Script know of geometry by option usegeometry=true.

```
\usepackage[left=2cm, right=2, top=3cm, bottom=4cm,
bindingoffset=1cm, includeheadfoot]{geometry}
```

- Auto-completion determines unspecified dimensions (under or over specified as well), here width and height of text (see I.9).
- Other options: paper=a4paper, landscape|portrait, includehead, includefoot, includeheadfoot, twocolumn
- Changing page layout mid document: \newgeometry{opt, opt, ...}

### C.2. Header and Footer of Page (aka running heading)



```
% delete default settings and define your own
\usepackage[automark]{scrlayer-scrpage}
\clearpaïrpagestyles
\ohead[]{\headmark} \ofoot[\pagemark]{\pagemark}
```

```
% Variant for a thesis with horizontal rules at head and foot
\usepackage[headsepline=0.005pt:,footsepline=0.005pt:,
plainfootsepline,automark]{scrlayer-scrpage}
\clearpaïrpagestyles
\ohead[]{\headmark} \ofoot[\pagemark]{\pagemark}
\ModifyLayer[addvoffset=-.6ex]{scrheadings.foot.above.line}
\ModifyLayer[addvoffset=-.6ex]{plain.scrheadings.foot.above.line}
\setkomafont{pageheadfoot}{\small}
```

### C.3. Linespread with setspace

Load: \usepackage[onehalfspacing]{setspace} for 1.5 line spacing.

## D. Document Structure (see also L)

### D.1. Start Document

\begin{document} Complete document contents. \end{document}

### D.2. Title

simple title: \author{text} \title{text} \date{\today} \maketitle  
title page self designed: \begin{titlepage} text \end{titlepage}

D.3. Table of Contents, List of Figures (for other List of see E.8 & G)  
\tableofcontents \listoftables \listoffigures  
KOMAoption toc=listof (see B.1) generates entries for TOC.

### D.4. Headings

| \part{title}      | \chapter{title}      |                       |
|-------------------|----------------------|-----------------------|
| \section{title}   | \subsection{title}   | \subsubsection{title} |
| \paragraph{title} | \subparagraph{title} |                       |

→ \chapter only valid in documentclass scrbook and scrreprt  
→ Use \* variants for headings without numbering, no change in counter and no entry in table of contents.  
→ Use the optional parameter for short titles in headings and table of contents, e.g. \section[short title]{title}  
→ Use \addpart, \addchap or \addsec for unnumbered headings, but with running heading and entry in table of contents. The \* variants delete the running heading.  
→ Layout of paragraph and subparagraph similar to other headings:  
\RedeclareSectionCommands[after skip=1sp]{paragraph, subparagraph}  
\setcounter{secnumdepth}{\subparagraphnumdepth}  
\setcounter{tocdepth}{\subparagraphnumdepth}

### D.5. Justification

| Environment        | Declaration  | Other                       |
|--------------------|--------------|-----------------------------|
| \begin{center}     | \centering   | text \par\vfill text        |
| \begin{flushleft}  | \raggedright | text \hfill text            |
| \begin{flushright} | \raggedleft  | \raggedbottom, \flushbottom |

### D.6. Lists

```
\begin{itemize} with bullets \item or \item[symbol]
\begin{enumerate} with numbers \item
\begin{description} with bold words \item[word]
\begin{labeling}[separator]{labelinglabel} \item[word]

\begin{enumerate}
\item First item
\item Second item\label{it:second} % see References
\end{enumerate}
```

### D.7. Enhanced Lists with enumitem

Load: \usepackage{enumitem}  
Example (for enumerate):  
\setlist[enumerate,1]{label=\Alph\*}} A) one  
\setlist[enumerate,2]{label=\alpha\*}} a) one  
\setlist[enumerate,3]{label=\roman\*}} b) two  
\setlist[enumerate,4]{label=\arabic\*}} B) two  
Example (for legal list):  
\newlist{legal}{enumerate}{10} 1. one  
\setlist{legal}{label=\arabic\*,.noitemsep} 1.1. two  
Use: \begin{legal} \item ... \end{legal} 1.1.1. three  
1.1.2. strawberry

### D.8. Separate Files

- After preamble within the text place: \include{file} Text starts and ends on a new page. file has to be in the same directory as the master file. Otherwise specify a path: \include{path/file}
- In preamble place: \includeonly{file1,file2} to run only these files.
- Use \input{file} includes a file without starting/ending on a new page (\includeonly not valid).

## E. Text

### E.1. Paragraphs (≈ "new idea in content")

Paragraphs are separated by an empty line in the code or by \par.  
A \ produces a new line – use sparingly, seldom needed outside tabulars.  
Correct Overfull Box Warnings with more than 4pt (look into log file).

### E.2. Text Symbols/Characters (see also A.2)

A lot of diacritic symbols can be typed directly, e.g. è é ê ñ ç  
§ \S \_ \textunderscore{} ~ \textasciitilde{}  
^ \textasciicircum{} ... \ldots | \textbar  
Other symbols need packages, e.g. € \texteuro (textcomp)

### E.3. Fonts

| Command       | Declaration      | Effect               |
|---------------|------------------|----------------------|
| \textrm{text} | {\rmfamily text} | Roman family         |
| \textsf{text} | {\sffamily text} | Sans serif family    |
| \texttt{text} | {\ttfamily text} | Typewriter family    |
| \textmd{text} | {\mdseries text} | Medium series        |
| \textbf{text} | {\bfseries text} | <b>Bold series</b>   |
| \textup{text} | {\upshape text}  | Upright shape        |
| \textit{text} | {\itshape text}  | <i>Italic shape</i>  |
| \textsl{text} | {\slshape text}  | <i>Slanted shape</i> |
| \textsc{text} | {\scshape text}  | SMALL CAPS SHAPE     |

More general commands:  
\emph{text} {\em text} *Emphasized*  
\textnormal{text} {\normalfont text} Document font

### E.4. Font Size

Font size is relative to the base font size, specified in the document class.

|               |                             |        |       |
|---------------|-----------------------------|--------|-------|
| \tiny         | <small>tiny</small>         | \Large | Large |
| \scriptsize   | <small>scriptsize</small>   | \LARGE | LARGE |
| \footnotesize | <small>footnotesize</small> |        |       |
| \small        | <small>small</small>        | \huge  | huge  |
| \normalsize   | <small>normalsize</small>   |        |       |
| \large        | <small>large</small>        | \Huge  | Huge  |

Use: {\small text} or {\huge text}\par to limit the size change.  
Example: \setkomafont{pageheadfoot}{\small}

### E.5. Colors with xcolor

```
\usepackage{xcolor}
\definecolor{DarkBlue}{RGB}{0, 115, 207}
\colorlet{col.section}{DarkBlue}
\textcolor{red}{text in red} or {\color{red}text}
\colorbox{gray!25}{\color{gray} faded by 25\%}
```

Predefined colors:

white gray black red green blue cyan magenta yellow

Fade a color with color!value between 0 and 100

Headings in color: \setkomafont{disposition}{\color{color}}

### E.6. Footnotes

|                     |                                                                |
|---------------------|----------------------------------------------------------------|
| \footnote{text}     | Print footnote marker in text and footnote at bottom of page   |
| \footnotemark       | Print footnote marker in text (e.g. within tabular or caption) |
| \footnotetext{text} | Print footnote at bottom of page                               |

### E.7. References with hyperref (loads url implicitly)

|                           |                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------|
| \autocite{citekey}        | Cite a bibliographic reference (package biblatex)                                         |
| \label{marker}            | Set a marker for cross reference, often if the form \label{sec:item} or \label{fig:diag1} |
| \autoref{marker}          | Give type name and number of marker                                                       |
| \autopageref{marker}      | Give abbreviation of "page" and page number of marker                                     |
| \url{url}                 | Print clickable web page                                                                  |
| \href[options]{url}{text} | Print clickable link                                                                      |
| \hyperref[marker]{text}   | Print clickable reference                                                                 |

Style: \urlstyle{xx} with xx a style like "tt", "rm", "sf" or "same".  
Names for autoref (package babel):  
\renewcaptionname{language}{\typenameautorefname}{text},  
e.g. \renewcaptionname{english}{\subsectionautorefname}{section}

### E.8. Acronyms with acro

```
\usepackage{acro,hyperref,longtable,tabu} \next 5 to praeambel
\acsetup{list-style=longtabu,list-heading=addchap}
\DeclareAcroListStyle{longtabu}{table}{table=longtabu,
table-spec=@{}{\bfseries}\X@{}}
\DeclareAcronym{ecm}{short=EM,long=Electro Machining}
...
\ac{EM} or \Ac{EM} for capitalized first letter
\printacronyms
```

## F. Figures & Tables (floating environments)

### F.1. Figures with graphicx

Load: \usepackage{graphicx}  
Use: \includegraphics[opt]{file} (png, jpg, pdf)  
With 'figure' the environment to place a graphic is meant. The figure caption is printed where the caption command is placed in the input. Extra vertical space is controlled by the KOMAoption captions (see B.1).  
Use: \begin{figure}[pos] ..\caption{..}\label{fig:x} \end{figure}  
Parameter: pos is a suggestion for placing, it can be ignored by T<sub>E</sub>X.  
Possible values are combinations of t (top), h (here), b (bottom), ! (try harder), p (separate page).  
Hint: Define a path to the graphic files (no blanks in folder names; no special characters in file names) \graphicspath{ {folder/}{folder/}... }

```
\graphicspath{ {img/} } %subfolder for images; set in preamble
\begin{figure}\centering
\includegraphics[width=.8\columnwidth]{pic.jpg}
\caption[Short title]{Long title}\label{fig:ff}
\end{figure}
```

- Numbering throughout the whole document (scrbook) with package chngcntr: \counterwithout{figure}{chapter} (same for table)
- Figure name: \renewcaptionname{language}{\figurename}{text}  
\renewcaptionname{language}{\figureautorefname}{text}





| <div></div> <div><div>My Title</div><div>My Subtitle</div><div>N.N.</div></div>                                                                                                                   | <div>Contents</div> <div><div>List of Figures</div><div>List of Tables</div><div>1. Intro</div><div>1.1. Technical introduction</div><div>1.1.1. Heading lower level</div><div>2. Method</div><div>A. Appendix</div><div>Bibliography</div></div> <div><div>iii</div><div>iv</div><div>1</div><div>1</div><div>1</div><div>2</div><div>4</div><div>5</div></div> | <div>List of Figures</div> <div><div>A.1. Example of a caption centeredbeside</div></div> <div><div>4</div></div>                           | <div>List of Tables</div> <div><div>1.1. Table caption</div></div> <div><div>1</div></div>                                                                                                                                      | <div>1. Introduction</div> <div><div>1.1. Technical introduction</div></div> <div>Heading without number</div> <div>Some text in <b>bold</b>, some more text. Some text in <i>emph</i>, more text. And now some text in German language Hier kommt eine Formel: <math>2 + 2 = 5</math></div> <div>A new paragraph, viz. a new idea, a new thought.</div> <div><div>1.1.1. Heading lower level</div></div> <div>More text <i>emphasized</i> text.<sup>1</sup></div> <div>Table 1.1.: Table caption</div> <div><table><tr><th>Head</th><th>Head</th><th>Head</th></tr><tr><td>Data</td><td>first</td><td>Row</td></tr><tr><td>Data</td><td>second</td><td>Row</td></tr></table></div> <div><div><sup>1</sup>Wikipedia. TeX - Wikipedia, The Free Encyclopedia. <a href="https://en.wikipedia.org/wiki/TeX">https://en.wikipedia.org/wiki/TeX</a>, last referenced 24-June-2017.</div><div>1</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Head | Head | Head | Data | first | Row | Data | second | Row |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|-------|-----|------|--------|-----|
| Head                                                                                                                                                                                                                                                                               | Head                                                                                                                                                                                                                                                                                                                                                             | Head                                                                                                                                        |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |      |      |      |       |     |      |        |     |
| Data                                                                                                                                                                                                                                                                               | first                                                                                                                                                                                                                                                                                                                                                            | Row                                                                                                                                         |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |      |      |      |       |     |      |        |     |
| Data                                                                                                                                                                                                                                                                               | second                                                                                                                                                                                                                                                                                                                                                           | Row                                                                                                                                         |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |      |      |      |       |     |      |        |     |
| <div>2. Method</div> <div>Some text, see Table 1.1 for aligned material.</div> <div>L'Hopital's rule:</div> <div>(2.1) <math>\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}}{\frac{1}{2}} = \frac{1}{2}</math></div> <div><div>2</div></div> | <div>2. Method</div> <div>More text. A "quote" and a „Zitat“.</div> <div><div>ii</div></div>                                                                                                                                                                                                                                                                     | <div>A. Appendix</div> <div><div>A BOX</div></div> <div>Figure A.1.: Example of a caption beside the figure</div> <div><div>iii</div></div> | <div>Bibliography</div> <div>Wikipedia. TeX - Wikipedia, The Free Encyclopedia. <a href="https://en.wikipedia.org/wiki/TeX">https://en.wikipedia.org/wiki/TeX</a>, last referenced 24-June-2017.</div> <div><div>iv</div></div> | <div>License</div> <div>Reference Sheet, Version 1.2, August 2017</div> <div>This Cheat Sheet is free under the terms and conditions of the L<sup>A</sup>T<sub>E</sub>X Project Public License Version 1.3c.</div> <div>Authors: Marion Lammarch, University of Heidelberg, and Elke Schubert, Stutenrose</div> <div>Credit for the layout and content ideas goes to</div> <div>(a) Emanuel Regnath, "L<sup>A</sup>T<sub>E</sub>X Cheat Sheet" L<sup>A</sup>T<sub>E</sub>X 2<sub>ε</sub> 2013<br/><a href="http://www.latexte.de/downloads/LaTeX_CheatSheet.pdf">http://www.latexte.de/downloads/LaTeX_CheatSheet.pdf</a></div> <div>(b) Tanno Schwidh, DHBW Mosbach, "LaTeX Cheat German", Ed. 2014<br/><a href="http://mirror.ctan.org/tex-archive/info/latexcheat/latexcheat-de/">http://mirror.ctan.org/tex-archive/info/latexcheat/latexcheat-de/</a></div> <div>The logo with the TeX lion is sketched by Dwayne Hilby, <a href="https://mirror.ctan.org/1306/t1306">https://mirror.ctan.org/1306/t1306</a>; the graphic of a header of a page is from the KOMA-Script documentation, file scrlayer-ocrpage.tex; the page layout image is done by using the package layout and the command \layout and the red lines and red text added with a graphics program.</div> <div>Acknowledgements</div> <div>We would like to thank Markus Kohn, Ulrike Fischer, Martin Sievers, and Herbert Voß for their useful hints and suggestions.</div> <div>Not naming all the female and male students which with their intrigued questions helped to improve this reference sheet.</div> <div><div>5</div></div> |      |      |      |      |       |     |      |        |     |